System Functional Requirement Specifications

1. The system shall be able to read and verify that the user selects a .txt file from their computer.

2. The system shall be able to interpret instructions within the file in the correct format supplied by the README document.

3. The system shall be able to store user input values into an accumulator for arithmetic operations

4. The system shall be able to perform addition, subtraction, multiplication, and division on user submitted values.

5. The system shall be able to store user submitted values into a location in memory starting from memory location 00 and incrementing.

6. The system shall be able to prompt the user for a integer value and determine if it is a valid integer.

7. The system shall be able to read a value from a specific memory location and display it to the user.

8. The system shall be able to read and understand a given instruction that is stored within a specific memory location.

9. The system shall be able to branch to a given memory location provided in the instruction file.

10. The system shall be able to display values stored in memory to the user given a specific memory location.

11. The system shall be able to display the current accumulator value as it changes based on instructions.

12. The system shall be able to branch to a specific memory location if the accumulator is a negative value.

13. The system shall be able to branch to a specific memory location if the accumulator value is zero.

14. The system shall be able to halt the program given a specific instruction value.

15. The system shall be able to handle overflow values by truncating the value to contain the last four digits.

System Non-functional Requirements

1. The system shall have a GUI that is colorblind accessible.

2. The system shall have the user login using a username and password credentials.

3. The system shall have a feedback button on the application to submit user feedback.